

Design of a Stable Fuzzy Controller for an Articulated Vehicle

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Abstract—This paper presents a backward movement control of an articulated vehicle via a model-based fuzzy control technique. A nonlinear dynamic model of the articulated vehicle is represented by a Takagi–Sugeno fuzzy model. The concept of parallel distributed compensation is employed to design a fuzzy controller from the Takagi–Sugeno fuzzy model of the articulated vehicle. Stability of the designed fuzzy control system is guaranteed via Lyapunov approach. The stability conditions are characterized in terms of linear matrix inequalities since the stability analysis is reduced to a problem of finding a common Lyapunov function for a set of Lyapunov inequalities. Simulation results and experimental results show that the designed fuzzy controller effectively achieves the backward movement control of the articulated vehicle.

I. INTRODUCTION

Backward movement control of articulated vehicles such as trailer type mobile robots is difficult since they are nonlinear and non-holonomic systems. To realize backward movement of articulated vehicles, we have to avoid “jackknife” phenomenon. Backward movement controls of computer simulated articulated vehicles have been realized via intelligent controls such as fuzzy control, neural control, and both of them [2], [3], [6], [8], [16]. However, stability of the control systems has not been analyzed in the papers. Recently, Sampei *et al.* [9] developed the backward movement control system of a real trailer type mobile robot using time scale transformation and exact linearization. This is a powerful design method for nonlinear systems if exact linearization is achieved. However, it requires special and rather involved knowledge. On the other hand, the design procedure discussed here is quite straightforward and natural.

This paper consists of four parts: fuzzy modeling of an articulated vehicle, fuzzy controller design via parallel distributed compensation, stability analysis of the designed control system and simulation and experimental results. Section II shows a nonlinear model of an articulated vehicle and fuzzy modeling for the nonlinear model. A nonlinear dynamic model of the articulated vehicle is represented by a Takagi–Sugeno fuzzy model. Section III shows the parallel distributed compensation. The parallel distributed compensation is employed to design a fuzzy controller from the Takagi–Sugeno fuzzy model of the articulated vehicle. Section IV describes stability analysis based on Lyapunov approach. The stability conditions are characterized in terms of linear matrix inequalities. Section V shows simulation and experimental results. The results show that the designed fuzzy controller effectively realizes the backward movement control even from difficult initial positions.

II. FUZZY MODELING OF AN ARTICULATED VEHICLE

Tokunaga and Ichihashi [16] formulated a nonlinear model of an articulated vehicle

$$x_0(t+1) = x_0(t) + \frac{v \cdot \Delta t}{l} \tan(u(t)) \quad (1)$$

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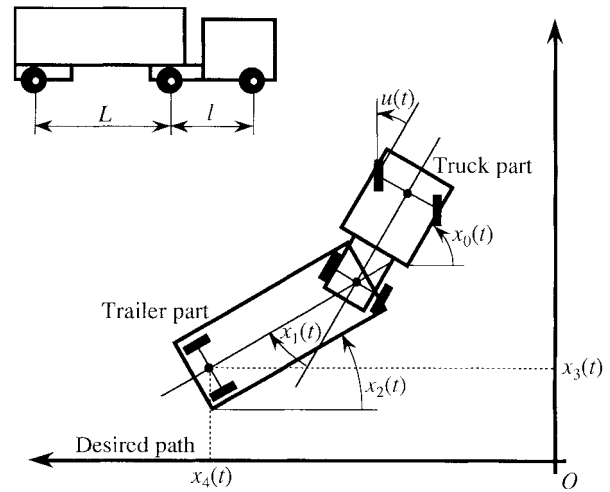


Fig. 1. Articulated vehicle.

$$x_1(t) = x_0(t) - x_2(t) \quad (2)$$

$$x_2(t+1) = x_2(t) + \frac{v \cdot \Delta t}{L} \sin(x_1(t)) \quad (3)$$

$$x_3(t+1) = x_3(t) + v \cdot \Delta t \cdot \cos(x_1(t)) \cdot \sin\left(\frac{x_2(t+1) + x_2(t)}{2}\right) \quad (4)$$

$$x_4(t+1) = x_4(t) + v \cdot \Delta t \cdot \cos(x_1(t)) \cdot \cos\left(\frac{x_2(t+1) + x_2(t)}{2}\right) \quad (5)$$

where

- $x_0(t)$ angle of truck;
- $x_1(t)$ angle difference between truck and trailer;
- $x_2(t)$ angle of trailer;
- $x_3(t)$ vertical position of rear end of trailer;
- $x_4(t)$ horizontal position of rear end of trailer;
- $u(t)$ steering angle.

In this experiment,

$$l = 0.20 \text{ [m]}, \quad L = 0.32 \text{ [m]},$$

$$v = -0.10 \text{ [m/s]}, \quad \Delta t = 0.5 \text{ [s]}.$$

l is the length of truck, L is the length of trailer, Δt is the sampling time, and v is the constant speed of the backward movement. With respect to $x_1(t)$, 90° and -90° correspond to two “jackknife” positions. Fig. 1 shows the articulated vehicle model and its coordinate system. The control purpose is to realize the backward movement of the articulated vehicle along the straight line ($x_3 = 0$), i.e.,

$$x_1(t) \rightarrow 0, \quad x_2(t) \rightarrow 0, \quad x_3(t) \rightarrow 0$$

without forward movements.

To realize model-based fuzzy controller design, we should make a fuzzy model which represents the dynamics of the articulated vehicle. After the nonlinear model of the articulated vehicle is simplified, an exact fuzzy modeling technique [5] is used to obtain the fuzzy model (6) from the simplified nonlinear model. The outline is presented in Appendix I.

$$\begin{aligned} \text{Rule 1: IF } x_2(t) + \frac{v \cdot \Delta t}{2L} x_1(t) \text{ is "about 0 [rad]} \\ \text{THEN } \mathbf{x}(t+1) = \mathbf{A}_1 \mathbf{x}(t) + \mathbf{b}_1 u(t) \end{aligned}$$

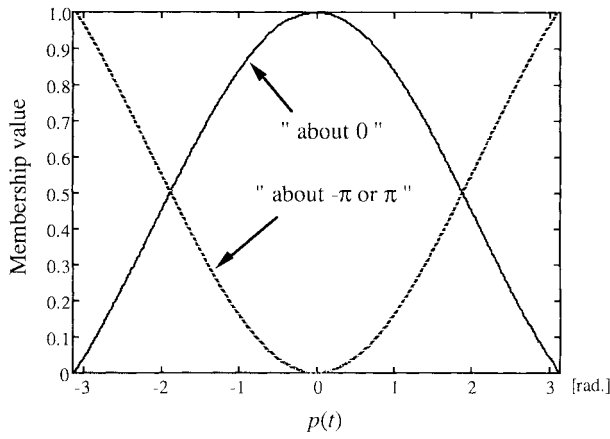


Fig. 2. Membership functions.

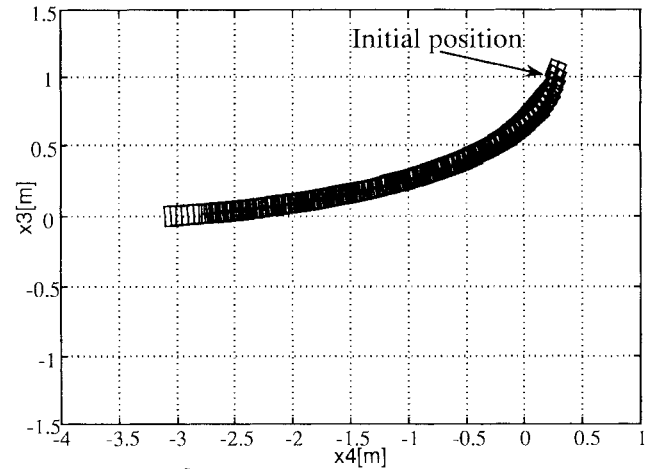


Fig. 5. Simulation result (original model, case 1).

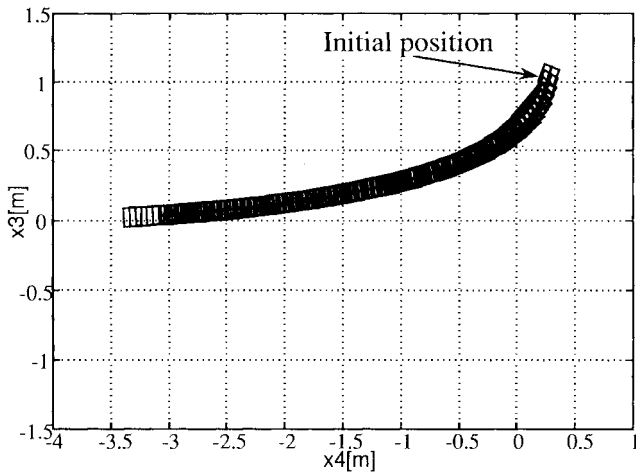


Fig. 3. Simulation result (fuzzy model, case 1).

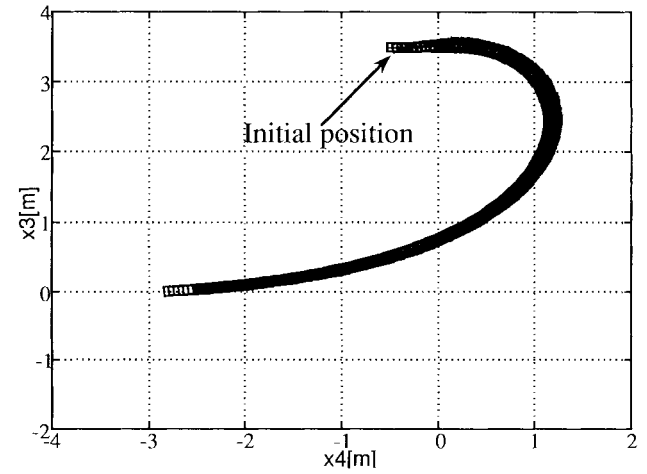


Fig. 6. Simulation result (original model, case 2).

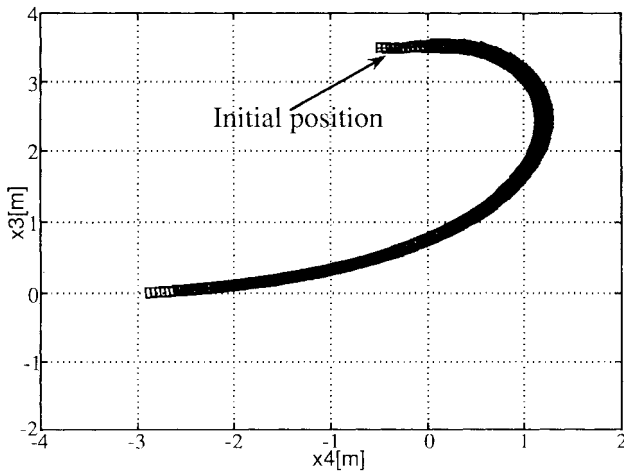


Fig. 4. Simulation result (fuzzy model, case 2).

$$\text{Rule 2: IF } x_2(t) + \frac{v \cdot \Delta t}{2L} x_1(t) \text{ is "about } \pi \text{ [rad]"} \quad (6)$$

$$\text{or } -\pi \text{ [rad]}"$$

$$\text{THEN } \mathbf{x}(t+1) = \mathbf{A}_2 \mathbf{x}(t) + \mathbf{b}_2 u(t)$$

where

$$\mathbf{x}(t) = [x_1(t) \quad x_2(t) \quad x_3(t)]^T$$

$$\mathbf{A}_1 = \begin{bmatrix} 1 - \frac{v \cdot \Delta t}{L} & 0 & 0 \\ \frac{v \cdot \Delta t}{L} & 1 & 0 \\ \frac{(v \cdot \Delta t)^2}{2L} & v \cdot \Delta t & 1 \end{bmatrix}$$

$$\mathbf{b}_1 = \begin{bmatrix} \frac{v \cdot \Delta t}{L} \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{A}_2 = \begin{bmatrix} 1 - \frac{v \cdot \Delta t}{L} & 0 & 0 \\ \frac{v \cdot \Delta t}{L} & 1 & 0 \\ \frac{g \cdot (v \cdot \Delta t)^2}{2L} & g \cdot v \cdot \Delta t & 1 \end{bmatrix}$$

$$\mathbf{b}_2 = \begin{bmatrix} \frac{v \cdot \Delta t}{L} \\ 0 \\ 0 \end{bmatrix}$$

$$g = \frac{10^{-2}}{\pi}$$

As pointed out in Appendix I, this fuzzy model exactly represents the simplified nonlinear model (A.1)–(A.3) under the condition of

$$-179.4270^\circ < x_2(t) + \frac{v \cdot \Delta t}{2L} x_1(t) < 179.4270^\circ.$$

The fuzzy model (6) is inferred as

$$\mathbf{x}(t+1) = \frac{\sum_{i=1}^2 w_i(p(t)) \{ \mathbf{A}_i \mathbf{x}(t) + \mathbf{b}_i u(t) \}}{\sum_{i=1}^2 w_i(p(t))} \quad (7)$$



Fig. 7. Photograph of articulated vehicle.

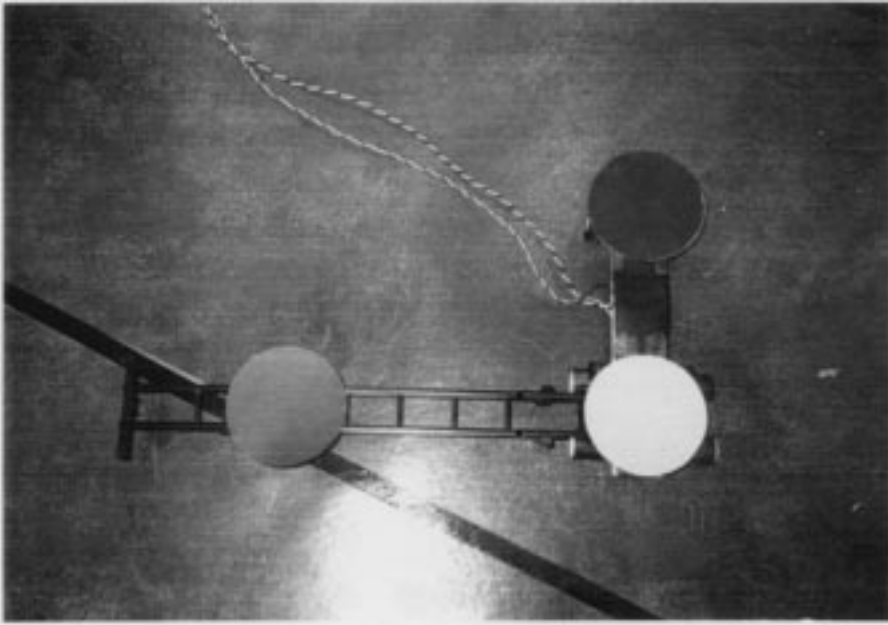


Fig. 8. Photograph of "jackknife" position.

where

$$w_1(p(t)) = \frac{\sin[p(t)] - g \cdot p(t)}{p(t) \cdot (1 - g)}$$

$$w_2(p(t)) = \frac{p(t) - \sin[p(t)]}{p(t) \cdot (1 - g)}$$

$$p(t) = x_2(t) + \frac{v \cdot \Delta t}{2L} \cdot x_1(t)$$

$$g = \frac{10^{-2}}{\pi}.$$

$w_i(p(t))$ is the membership value of the fuzzy set in Rule i . Fig. 2 shows the fuzzy sets of "about 0 [rad]" and "about π [rad] or $-\pi$ [rad]."

III. FUZZY CONTROLLER DESIGN VIA PARALLEL DISTRIBUTED COMPENSATION

Section III presents a fuzzy controller design via the parallel distributed compensation. The main idea of the parallel distributed compensation is to derive each control rule so as to compensate each rule of a fuzzy system [12], [17], [18]. Control Rule 1 and Control Rule 2 of the fuzzy controller can be designed from Rule 1 and Rule 2 of the fuzzy model (6). We use the exact same fuzzy sets in the design of the premise parts of the fuzzy controller.

Control Rule 1:

$$\text{IF } x_2(t) + \frac{v \cdot \Delta t}{2L} x_1(t) \text{ is "about 0 [rad]"} \\ \text{THEN } u(t) = \mathbf{f}_1 \mathbf{x}(t).$$

Control Rule 2:

$$\text{IF } x_2(t) + \frac{v \cdot \Delta t}{2L} x_1(t) \text{ is "about } \pi \text{ [rad]"} \\ \text{or } -\pi \text{ [rad]"} \\ \text{THEN } u(t) = \mathbf{f}_2 \mathbf{x}(t)$$

where

$$\mathbf{f}_1 = [f_{11} \ f_{12} \ f_{13}] \\ \mathbf{f}_2 = [f_{21} \ f_{22} \ f_{23}].$$

The fuzzy controller (8) is represented as

$$u(t) = \frac{\sum_{i=1}^2 w_i(p(t)) \mathbf{f}_i \mathbf{x}(t)}{\sum_{i=1}^2 w_i(p(t))}. \quad (9)$$

The purpose of controller design is to determine the feedback gains $\mathbf{f}_1$ and $\mathbf{f}_2$. The Riccati equation for linear discrete systems was employed to determine these feedback gains since each consequent part is represented by a linear state equation. The feedback gains will be presented in Appendix II.

IV. STABILITY ANALYSIS

Section IV presents stability analysis of the designed control system. By substituting (9) into (7), we obtain

$$\mathbf{x}(t+1) = \frac{\sum_{i=1}^2 w_i(p(t)) w_j(p(t)) \{\mathbf{A}_i + \mathbf{b}_i \mathbf{f}_i\} \mathbf{x}(t)}{\sum_{i=1}^2 \sum_{j=1}^2 w_i(p(t)) w_j(p(t))} \\ + \frac{\sum_{i < j} 2 w_i(p(t)) w_j(p(t)) \frac{\{\mathbf{A}_i + \mathbf{b}_i \mathbf{f}_i\} + \{\mathbf{A}_j + \mathbf{b}_j \mathbf{f}_j\}}{2} \mathbf{x}(t)}{\sum_{i=1}^2 \sum_{j=1}^2 w_i(p(t)) w_j(p(t))}. \quad (10)$$

We derive the following stability conditions from (10) and Theorem A.1. Theorem A.1 will be given in Appendix III.

Theorem 4.1: The equilibrium of the fuzzy control system described by (10) is asymptotically stable in the large if there exists a common positive definite matrix $\mathbf{P}$ such that

$$\{\mathbf{A}_i + \mathbf{b}_i \mathbf{f}_i\}^T \mathbf{P} \{\mathbf{A}_i + \mathbf{b}_i \mathbf{f}_i\} - \mathbf{P} < 0 \quad (11)$$

$$\mathbf{G}_{ij}^T \mathbf{P} \mathbf{G}_{ij} - \mathbf{P} < 0, \quad i < j \quad (12)$$

for $i, j = 1, 2$, where

$$\mathbf{G}_{ij} = \frac{\{\mathbf{A}_i + \mathbf{b}_i \mathbf{f}_i\} + \{\mathbf{A}_j + \mathbf{b}_j \mathbf{f}_j\}}{2}. \quad (13)$$

Remark 4.1: To check stability of the fuzzy control system, it has long been considered difficult to find a common positive definite matrix $\mathbf{P}$ satisfying the conditions of (11) and (12). Most of the time, a trial-and-error type of procedure is used [11]–[13]. In [4], a procedure to construct a common $\mathbf{P}$ is given for second order fuzzy systems, i.e., the dimension of the state is two. We pointed out in [17]–[19] that the common $\mathbf{P}$ problem for fuzzy control systems can be solved numerically, i.e., the stability condition of Theorem 4.1 can be expressed in linear matrix inequalities (LMI's) [1]. To check stability, we need to find $\mathbf{P}$ satisfying the LMI

$$\mathbf{P} > 0, \\ \{\mathbf{A}_i + \mathbf{b}_i \mathbf{f}_i\}^T \mathbf{P} \{\mathbf{A}_i + \mathbf{b}_i \mathbf{f}_i\} - \mathbf{P} < 0, \\ \mathbf{G}_{ij}^T \mathbf{P} \mathbf{G}_{ij} - \mathbf{P} < 0, \quad i < j$$

or determination that no such $\mathbf{P}$ exists. This is a convex feasibility problem. Numerically, this feasibility problem can be solved very efficiently by means of the most powerful tools available to date in the mathematical programming literature. For instance, recently developed interior-point methods [7] are extremely efficient in practice. The details for an LMI approach to fuzzy control systems are presented in [19].

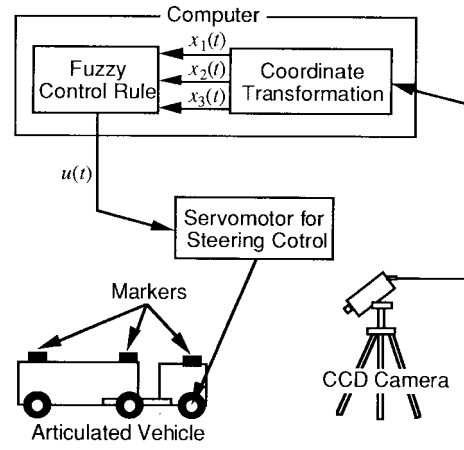


Fig. 9. Experimental system.

Remark 4.2: In the fuzzy model (7), $\mathbf{b}_1 = \mathbf{b}_2$. In this case, the stability conditions of Theorem 4.1 can be simplified as follows. The equilibrium of the fuzzy control system is asymptotically stable in the large if there exists a common positive definite matrix $\mathbf{P}$ satisfying (11). In this case, $\mathbf{G}_{ij}^T \mathbf{P} \mathbf{G}_{ij} - \mathbf{P} < 0$ if $\{\mathbf{A}_i + \mathbf{b}_i \mathbf{f}_i\}^T \mathbf{P} \{\mathbf{A}_i + \mathbf{b}_i \mathbf{f}_i\} - \mathbf{P} < 0$.

As shown in Appendix II, the feedback gains $\mathbf{f}_i$ are determined by solving the Riccati equation

$$\mathbf{f}_1 = [1.314, -0.138, 0.027] \\ \mathbf{f}_2 = [1.264, -0.085, 0.003].$$

If we choose

$$\mathbf{P} = \begin{bmatrix} 47.42 & -2904 & 114.8 \\ -2904 & 4172 & -158.2 \\ 114.8 & -158.2 & 529.1 \end{bmatrix} > 0$$

as a common positive definite matrix $\mathbf{P}$, then

$$\{\mathbf{A}_1 + \mathbf{b} \mathbf{f}_1\}^T \mathbf{P} \{\mathbf{A}_1 + \mathbf{b} \mathbf{f}_1\} - \mathbf{P} \\ = \begin{bmatrix} -638.2 & -7.537 & -22.37 \\ -7.537 & -178.0 & -4.066 \\ -22.37 & -4.066 & -1.327 \end{bmatrix} < 0 \\ \{\mathbf{A}_2 + \mathbf{b} \mathbf{f}_2\}^T \mathbf{P} \{\mathbf{A}_2 + \mathbf{b} \mathbf{f}_2\} - \mathbf{P} \\ = \begin{bmatrix} -528.5 & -93.54 & 3.403 \\ -93.54 & -121.9 & 4.261 \\ 3.403 & 4.261 & -0.153 \end{bmatrix} < 0.$$

Therefore, the designed control system is stable under the condition of

$$-179.4270^\circ < x_2(t) + \frac{v \cdot \Delta t}{2L} x_1(t) < 179.4270^\circ. \quad (14)$$

In other words, the backward movement control can be effectively achieved from all initial positions under the condition of (14). The common $\mathbf{P}$ is found by the LMI technique mentioned in Remark 4.1. Note that the common $\mathbf{P}$ satisfies

$$\mathbf{G}_{12}^T \mathbf{P} \mathbf{G}_{12} - \mathbf{P} = \begin{bmatrix} -583.6 & -50.34 & -9.574 \\ -50.34 & -150.4 & 0.183 \\ -9.574 & 0.183 & -0.783 \end{bmatrix} < 0.$$

Strictly speaking, the designed fuzzy controller guarantees the stability only for the simplified nonlinear model (A.1)–(A.3) [i.e., the fuzzy model (7)] derived in Appendix I. A future work is to guarantee stability not only for the simplified model but also for the original model (1)–(5).



(a)



(e)



(b)



(f)



(c)



(g)



(d)



(h)

Fig. 10. Experimental result 1 (photograph).

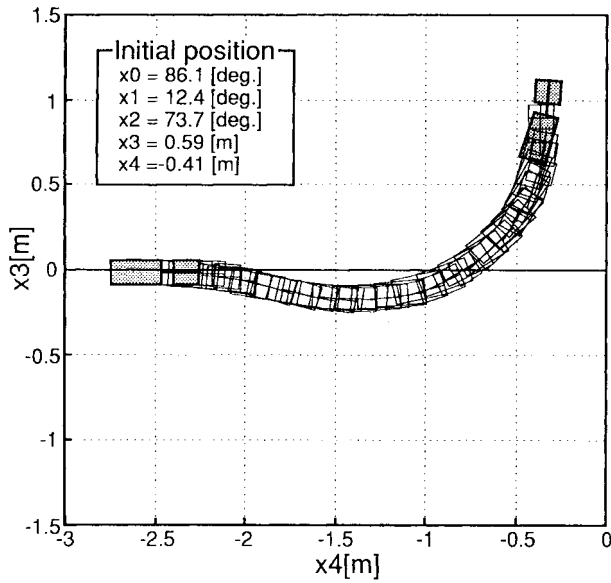


Fig. 11. Experimental result 1.

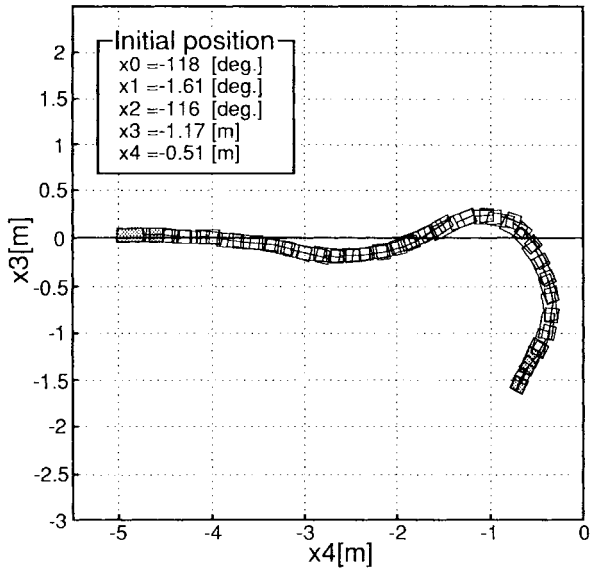


Fig. 12. Experimental result 2.

V. SIMULATION RESULTS AND EXPERIMENTAL RESULTS

The original model (1)–(5) and the fuzzy model (7) [i.e., the simplified nonlinear model (A.1)–(A.3)] are used in the simulation. Two cases of initial positions are selected in this simulation. Both of them requires the articulated vehicle to be turned in order to realize the backward movement control.

Figs. 3 and 4 show the simulation results for the fuzzy model (7). Figs. 5 and 6 show the simulation results for the original model (1)–(5). The following points can be made from the simulation results.

- 1) The designed fuzzy controller can effectively achieve the backward movement control even from difficult initial positions such as case 2 in both the original model and the fuzzy model (7).
- 2) The dynamics of the fuzzy model, i.e., the simplified nonlinear model, agrees well with that of the original model (1)–(5).

The first point is related to guarantee of the stability of the control system under the condition of (14). The second point shows that a

fairly good approximation is realized.

Fig. 7 shows the photograph of the articulated vehicle. In this experiment

$$l = 0.20 \text{ [m]}, L = 0.32 \text{ [m]}$$

$$v = -0.10 \text{ [m/s]}, \Delta t = 0.5 \text{ [s]}.$$

Fig. 8 shows a “jackknife” position. Fig. 9 shows the experimental system. Three markers are attached on the articulated vehicle. After measuring the positions of the markers using a CCD camera, the state variables $x_1(t)$, $x_2(t)$ and $x_3(t)$ are calculated and the steering angle $u(t)$ is determined. Figs. 10–12 show the control results for two cases of initial positions. It is found from the figures that the backward movement control of the articulated vehicle can be effectively realized by the fuzzy control. We emphasize that the nonlinear controller designed in [9] works only for $|x_2(t)| < 90^\circ$. Therefore, the nonlinear controller in [9] does not work for the initial position of the experimental result shown in Fig. 12.

VI. CONCLUSION

This paper has presented a backward movement control of an articulated vehicle via a model-based fuzzy control technique. A nonlinear dynamic model of the articulated vehicle has been represented by a Takagi–Sugeno fuzzy model. The concept of parallel distributed compensation has been employed to design a fuzzy controller from the Takagi–Sugeno fuzzy model of the articulated vehicle. Stability of the designed fuzzy control system has been guaranteed via Lyapunov approach. The stability conditions have been characterized in terms of linear matrix inequalities since the stability analysis is reduced to a problem of finding a common Lyapunov function for a set of Lyapunov inequalities. Simulation results and experimental results have shown that the designed fuzzy controller effectively achieves the backward movement control of the articulated vehicle.

APPENDIX I FUZZY MODELING

Assume that $u(t)$ and $x_1(t)$ are always small values for all t . Then, the original model (1)–(4) are simplified as

$$x_1(t+1) = \left(1 - \frac{v \cdot \Delta t}{L}\right) \cdot x_1(t) + \frac{v \cdot \Delta t}{l} \cdot u(t) \quad (\text{A.1})$$

$$x_2(t+1) = x_2(t) + \frac{v \cdot \Delta t}{L} \cdot x_1(t) \quad (\text{A.2})$$

$$x_3(t+1) = v \cdot \Delta t \cdot \sin\left(x_2(t) + \frac{v \cdot \Delta t \cdot x_1(t)}{2L}\right) + x_3(t). \quad (\text{A.3})$$

Notice that (A.3) has nonlinear term

$$v \cdot \Delta t \cdot \sin\left(x_2(t) + \frac{v \cdot \Delta t \cdot x_1(t)}{2L}\right). \quad (\text{A.4})$$

Equation (A.4) can be rewritten as

$$\begin{aligned} & \sin\left(x_2(t) + \frac{v \cdot \Delta t \cdot x_1(t)}{2L}\right) \\ &= w_1(p(t)) \cdot \left(x_2(t) + \frac{v \cdot \Delta t \cdot x_1(t)}{2L}\right) \\ &+ w_2(p(t)) \cdot g \cdot \left(x_2(t) + \frac{v \cdot \Delta t \cdot x_1(t)}{2L}\right) \end{aligned} \quad (\text{A.5})$$

where

$$g = \frac{10^{-2}}{\pi}$$

$$w_1(p(t)) = \frac{\sin[p(t)] - g \cdot p(t)}{p(t) \cdot (1 - g)}$$

$$w_2(p(t)) = \frac{p(t) - \sin[p(t)]}{p(t) \cdot (1 - g)}$$

$$p(t) = x_2(t) + \frac{v \cdot \Delta t}{2L} \cdot x_1(t).$$

Equation (A.5) exactly represents (A.4) under the condition of

$$-179.4270^\circ < p(t) < 179.4270^\circ.$$

If $w_1(p(t))$ and $w_2(p(t))$ are regarded as the membership values of “about 0 [rad]” and “about π [rad] or $-\pi$ [rad],” we obtain the fuzzy model (7). For more details, see [14].

APPENDIX II FEEDBACK GAINS

The following performance index is used for the determination of the feedback gains

$$J = \sum_{i=0}^{\infty} \mathbf{x}^T(t) \mathbf{Q} \mathbf{x}(t) + R u^2(t)$$

where

$$\text{Control rule 1 } \mathbf{Q} = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R = 1000$$

$$\text{Control rule 2 } \mathbf{Q} = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 0.01 \end{bmatrix}, \quad R = 1000.$$

By solving the Riccati equation, we obtain

$$f_{11} = 1.314, \quad f_{12} = -0.138, \quad f_{13} = 0.027$$

$$f_{21} = 1.264, \quad f_{22} = -0.085, \quad f_{23} = 0.003.$$

APPENDIX III STABILITY CONDITIONS

The open-loop system of the Takagi–Sugeno fuzzy system (7) is defined as

$$\mathbf{x}(t+1) = \frac{\sum_{i=1}^r w_i(p(t)) \mathbf{A}_i \mathbf{x}(t)}{\sum_{i=1}^r w_i(p(t))} \quad (\text{A.6})$$

where r is the number of if-then rules. Assume that

$$\sum_{i=1}^r w_i(p(t)) > 0,$$

$$w_i(p(t)) \geq 0, \quad i = 1, 2, \dots, r$$

for all t . Each linear consequent equation represented by $\mathbf{A}_i \mathbf{x}(t)$ is called “subsystem.”

A stability condition [11], [12] is derived for ensuring stability of (A.6) in accordance with the definition of stability in the sense of Lyapunov. Theorem A.1 gives the stability condition.

Theorem A.1: The equilibrium of the fuzzy system described by (A.6) is asymptotically stable in the large if there exists a common positive definite matrix $\mathbf{P}$ such that

$$\mathbf{A}_i^T \mathbf{P} \mathbf{A}_i - \mathbf{P} < \mathbf{0} \quad (\text{A.7})$$

for $i = 1, 2, \dots, r$, that is, for all the subsystems.

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REFERENCES

- [1] S. Boyd *et al.*, *Linear Matrix Inequalities in Systems and Control Theory*. Philadelphia, PA: SIAM, 1994.
- [2] H. Inoue, K. Kamei, and K. Inoue, “Auto-generation of fuzzy production rules using hyper-cone membership function by genetic algorithm,” in *Proc. Int. Joint Conf. CFS/IFIS/SOFT’95*, 1995, pp. 53–58.
- [3] R. E. Jenkins and B. P. Yuhas, “A simplified neural network solution through problem decomposition: The case of the truck backer-upper,” *IEEE Trans. Neural Networks*, vol. 4, no. 4, pp. 718–720, 1993.
- [4] S. Kawamoto, “An approach to stability analysis of second order fuzzy systems,” in *Proc. 1st IEEE Int. Conf. Fuzzy Systems*, 1992, pp. 1427–1434.
- [5] S. Kawamoto *et al.*, “Fuzzy-type Lyapunov function,” in *Proc. 2nd IEEE Int. Conf. Fuzzy Systems*, 1993.
- [6] S. G. Kong and B. Kosko, “Adaptive fuzzy systems for backing up a truck-and-trailer,” *IEEE Trans. Neural Networks*, vol. 3, pp. 211–223, Mar. 1992.
- [7] Y. Nesterov and A. Nemirovskii, *Interior-Point Polynomial Algorithms in Convex Programming*. Philadelphia, PA: SIAM, 1994.
- [8] D. Nguyen and B. Widrow, “The truck backer-upper: An example of self-learning in neural networks,” in *Proc. Int. Joint Conf. Neural Networks (IJCNN-89)*, June 1989, vol. 2, pp. 357–363.
- [9] M. Sampei *et al.*, “Arbitrary path tracking control of articulated vehicles using nonlinear control theory,” *IEEE Trans. Contr. Syst. Technol.*, vol. 3, no. 1, pp. 125–131, 1995.
- [10] T. Takagi and M. Sugeno, “Fuzzy identification of systems and its applications to modeling and control,” *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-15, no. 1, pp. 116–132, 1985.
- [11] K. Tanaka and M. Sugeno, “Stability analysis of fuzzy systems using Lyapunov’s direct method,” in *Proc. North American Fuzzy Information Processing Soc. (NAFIPS’90)*, 1990, pp. 133–136.
- [12] K. Tanaka and M. Sugeno, “Stability analysis and design of fuzzy control systems,” *Fuzzy Sets Syst.*, vol. 45, no. 2, pp. 135–156, 1992.
- [13] K. Tanaka and M. Sano, “Fuzzy stability criterion of a class of nonlinear systems,” *Inform. Sci.*, vol. 71, no. 1–2, Mar. 1993.
- [14] —, “A robust stabilization problem of fuzzy controller systems and its applications to backing up control of a truck-trailer,” *IEEE Trans. Fuzzy Syst.*, vol. 2, pp. 119–134, May 1994.
- [15] K. Tanaka, T. Ikeda, and H. Wang, “Robust stabilization of a class of uncertain nonlinear systems via fuzzy control: Quadratic stabilizability, H_∞ control theory and linear matrix inequalities,” *IEEE Trans. Fuzzy Syst.*, vol. 4, pp. 1–13, Feb. 1996.
- [16] M. Tokunaga and H. Ichihashi, “Backer-upper control of a trailer truck by neuro-fuzzy optimal control,” in *Proc. 8th Fuzzy System Symp.*, May 1992, pp. 49–52, in Japanese.
- [17] H. Wang, K. Tanaka, and M. Griffin, “Parallel distributed compensation of fuzzy systems via Takagi-Sugeno fuzzy model,” in *Proc. 4th IEEE Int. Conf. Fuzzy Systems*, 1995, vol. 2, pp. 531–538.
- [18] —, “An approach to fuzzy control of nonlinear systems,” unpublished.
- [19] H. Wang and K. Tanaka, “An LMI-based stable fuzzy control of nonlinear systems and its application to control of chaos,” in *IEEE 5th Int. Conf. Fuzzy Systems*, 1996, vol. 2, pp. 1433–1438.